

has always been discouraged on the premise that a rocky mountain desert cannot hold ground water. But, Arya punched holes in this belief by digging a borewell for the army at 14,000 ft in Chushul on China border in 2006—a feat that earned him an entry in the Guinness Book of World Records. "It's like rediscovering the simple principles of geology and physics operating in high altitudes," says Arya.

"His borewells are not only a cost-effective solution to the army's rising water needs in Ladakh but have also boosted the morale of the troops," says Samjay Kaul, assistant commander

works engineer at the newly-set up 14 Corps in Leh. "A systematic development of untapped potential of ground water can lead to green revolution in this cold desert," says Arya. He has since drilled borewells to augment water supply schemes for, among others, the Airports Authority of India, the Indian Oil Corporation, the Indian Air Force and field research laboratory of the Defence Research and Development Organisation. Ashok Sahni, Professor Emeritus at the Centre for Advanced Geology of Panjab University, calls Arya "an unconventional hydrogeologist". "He took a risk by drilling in the mountains and has struck abundant ground water where it was earlier seen as impossible,"

THE MOUNTAINMAN: Arya with a rig in the backdrop

says Sahni. Arya's discovery is based on practical experience gleaned from five years of digging hand pumps in Himachal Pradesh where he worked as a daily-wager hydrogeologist with the state government. What, however, added depth to his knowledge was his study of wells in 17th century forts on hill tops in Hamirpur and Solan districts. "The traditional mountain water supply techniques were based on intuitive science," says Arya.

To explain the prevalence of ground water in mountains, including the ones that have no rain-fed seepage or snow-fall, Arya divided the Himalayas into seven hydrostratigraphic zones in 1996 in a study he presented the same year at the International Conference of Geology in China. At the heart of his ground-breaking thesis is the finding that ground water resources in the Himalayan region depend on the type of rocks and structural parameters like folds, fissures and fault-lines in the rock strata. The mountains have ground aquifers just like plains but the water movement in high-altitudes is controlled by the principles of gravity and iso-stasy (wherein the water level is itself up). In his reckoning, even a barren mountain top below the height of Mount Everest will have ground water resources, provided it has favourable lithological conditions like the presence of water-absorbing sedimentary rocks or impervious rocks with water-trapping fault-lines. "Arya's high success rate in high-altitude borewells has flowed from his intimate technical knowledge of hydrogeology," says former CGWB chairman R.K. Chadda.

Arya has a near 100 per cent success rate in his borewells commissioned on a no-water-no-payment basis. Experts view his explorations as significant in the backdrop of receding glaciers and rising demand for water in the Ladakh region. Also, troop deployment, which has increased manifold since a new Corp was set up in Ladakh after the Kargil war in 1999, adds to the shortage. "The water level of the river Indus has fallen alarmingly this year, affecting irrigation schemes," says Chering Dorjay, Chairman of Ladakh Autonomous Hill Development Council.

Clearly, with Arya around, at least water is something that this cold desert will never thirst for. ■

ATER MAN



BANQUEE SINGH